

Capture the flag



PAGED OUT

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What is Paged Out!?

Paged Out! is a free experimental (one article == one page) technical magazine about programming (especially programming tricks!), [hacking](#), [security hacking](#), retro computers, modern computers, electronics, demoscene, and other similar topics.

It's **made by the community for the community**. And it's not-for-profit (though in time, we hope it will be self-sustained) - this means that the issues will always be free to download, share, and print.

If you're interested in more details, check our our [FAQ](#) and [About](#) pages!

Like what you see? See how [you can help us succeed!](#)

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Cover art by [ReFiend](#).

Issue #4: The epic Paged Out! story continues

- [Paged Out! #4 \(web PDF\)](#) (14MB)
- [Wallpaper \(JPG, 8K+, light version\)](#) (10MB)
- [Wallpaper \(PNG, 8K+, dark version\)](#) (43MB)

Note: This is a "beta build" of the PDF, i.e. we will be re-publishing it with various improvements multiple times. What's missing:

- [PDFs for printing \(A4+bleed\)](#) - we still need to fix the pipeline around this; will come out later

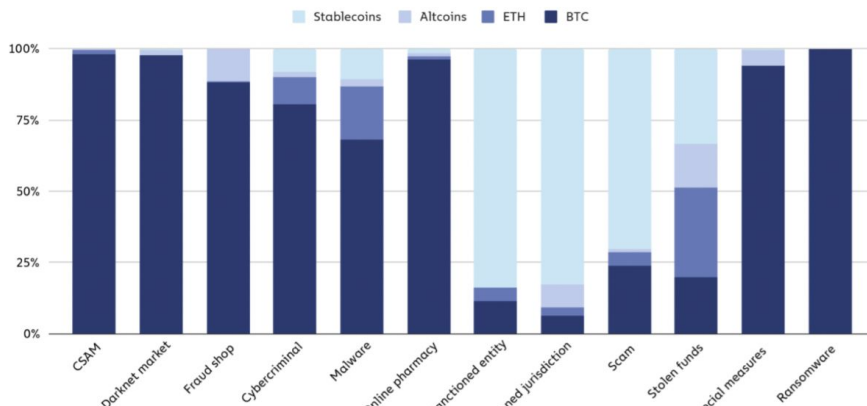
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What are ransomware groups doing with stolen money?

The rise of cryptocurrencies opened new doors for cybercriminals. Cryptocurrencies were designed to give the power over the money to the people as they are based on a decentralized network where no central authority controls them, provide a high level of anonymity, and it is much easier to move them across the border using e.g. cold wallets (hardware wallets). This is something that heavily attracts cybercriminals as well as the idea to obscure transactions and make them more difficult to track by law enforcement. With the growing cryptocurrency market, we've also seen an increased number of ransom payments received from attacked companies. Bitcoin is called the currency of the Dark Web as it's the crypto most often used by cybercriminals these days, especially in the Darknet market and ransomware.

But what does Dark Web actually mean? Dark Web is a service running over the Onion protocol which runs over the Tor network. It's a hidden part of the Internet that provides a high level of anonymity and privacy that may be used for good and bad reasons. The good one is e.g. escape from a country's censorship and the bad one is the already described market where criminals are selling and buying illegal services and products.

Illicit transaction volume by crime category and asset type, 2023



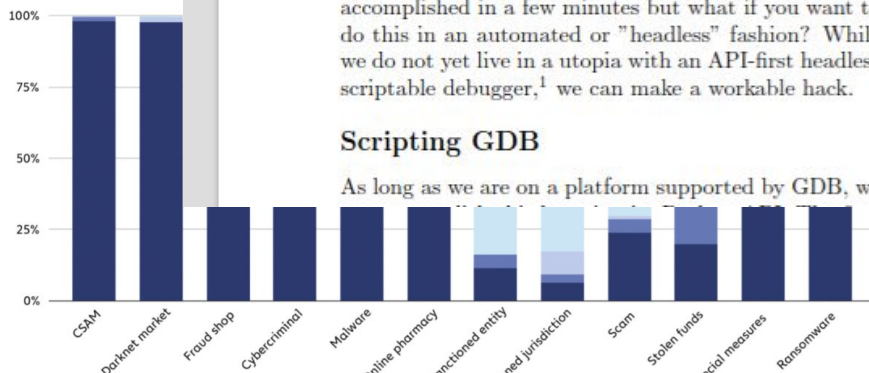
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What are r

The rise of cryptocurrencies money to the people as the level of anonymity, and it is something that heavily attracts by law enforcement. With the received from attacked companies by cybercriminals these days.

But what does Dark Web network. It's a hidden part of reasons. The good one is criminals are selling and buying

Illicit



Headless GDB Scripting

Consider a simple exploitation scenario where we have a stack buffer overflow and we want to figure out the location in our input that contains the value that will overwrite the return address. We could use our decompiler platform of choice, cross-reference the local variable usage, let it reconstruct the stack frame and compare the offset of the overflowing variable with the offset of the saved return address.

On the other hand, sending a de Bruijn sequence generated with the pwntools cyclic() function, using the instruction pointer value at the crash and calculating the offset, might be easier.

Doing this manually is all well and good and can be accomplished in a few minutes but what if you want to do this in an automated or "headless" fashion? While we do not yet live in a utopia with an API-first headless scriptable debugger,¹ we can make a workable hack.

Scripting GDB

As long as we are on a platform supported by GDB, we

```
class GdbExtractor:
    def __init__(self, gdb):
        self.semaphore = threading.Semaphore(0)
        self.gdb = gdb
        self.retaddr = None

    def __call__(self, event):
        self.retaddr = self.gdb...
        self.semaphore.release()

...
extractor = GdbExtractor(gdb_api)
gdb_api.events.stop.connect(extractor)
gdb_api.execute("continue")
...
extractor.semaphore.acquire()
print(f'Ret: {extractor.retaddr}')
```

For this to work fully headless, you might need to adjust the terminal. If you are using pwntools, you can do this like this:

```
context.terminal = [
```

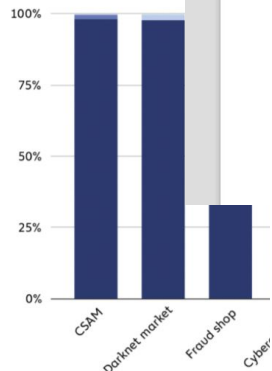
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Illici



Let Your Server Answer the Phone

Ft. Clvrpn, <https://disconnectmy.icu>

Have you ever wanted to play a game with a landline telephone? Do you need to chat with friends but Discord is down again? Did something happen to your VPS but you don't have a laptop nearby? Stop worrying and start using a PBX to solve your issues! A Private Branch Exchange, as official as it sounds, isn't just for businesses. With the proper configuration, your PBX can be used for fun as well! Here are a few ideas and example snippets to get started.

Note: The following examples are excerpts from dial plans, running on open source software Asterisk PBX (www.asterisk.org). You will also need a SIP trunking provider to forward VoIP calls to your server!

Dialplan Basics

In Asterisk, a dialplan is a file (or files) that determines how calls are answered, routed, and hung up as they pass through your PBX. Each entry in a dialplan has an extension, priority, and an application, such as:

```
exten => 300, 1, Answer(50)
```

This example simply opens a line when a call comes through on extension 300. Additional functions must have an incremental priority number. In general, be sure to always answer and hang up a call as it progresses through the dialplan. Start out with a simple dial plan that plays a greeting sound and waits 10 seconds for an extension:

```
exten=> _+<PBXnumber>,1,Answer(50)
same=> n,Background(greeting)
```

idea unless you are confident in your ability to configure one! The command will execute using the system's shell.

```
exten => 3,1,Shell(echo "Incoming!")
```

Dungeon Crawler

Stitching together extensions, redirects, and messages, it is easy to create a dungeon crawler style game playable entirely through a phone! For example, let buttons 2 and 8 be North/South and 4 and 6 be East/West.

```
[enter]
exten=>100,1,Background(greeting)
exten=>2,1,goto(locA,start,1) ;North
exten=>8,1,Playback(blocked) ;South (X)
exten=>8,2,goto(enter,100,1)

;East/West
exten=>4,1,Playback(blocked) ;East (X)
exten=>4,2,goto(enter,100,1)
exten=>6,1,goto(locD,start,1) ;West

[locA]
exten=>start,1,Playback(locA_audio)
exten=>8,1,goto(enter,100,1) ;South
exten=>_X,1,Playback(blocked)
same=>n,goto(locA)

[locD]
exten=>start,1,Playback(locD_audio)
exten=>6,1,goto(enter,100,1) ;West
exten=>_X,1,Playback(blocked)
same=>n,goto(locD)
```

This example only has three distinct areas, the entry area, location A (to the north) and location D (to the east). Attempting to move in a direction where there isn't an area will play a message claiming that the direction is

Quinesnake uses this string formatting to allow it to print its own source, but makes things more interesting by playing the classic game *snake* over the source (with wasd controls) after it's printed! It's

TODAY'S CHALLS

- 2 pwns
- 1 rev
- 2 webs
- 1 misc
- 2 crypto

chall.polygl0ts.ch

- 2 pwns
- 1 rev
- 2 webs
- 1 misc
- 2 crypto

Register

User Name

Your username on the site

Email

Never shown to the public

Password

Password used to log into your account

Submit

chall.polygl0ts.ch

- 2 pwns
- 2 rev
- 2 webs
- 1 misc
- 2 crypto

The screenshot shows the 'Challenges' page of the 'polygl0ts friday meetings' website. The page has a red header with navigation links: 'polygl0ts friday meetings', 'Users', 'Scoreboard', 'Challenges', 'Admin Panel', and 'Notifications'. The main content area is white and displays challenges categorized by type: 'rev', 'pwn', and 'crypto'. Each category contains two challenge cards, each with a title and a value of 500.

Category	Challenge Title	Value
rev	random means secure	500
	secure flag checker	500
pwn	are you admin?	500
	secure feedback	500
crypto	Really Secure Algorithm	500
	random really means secure	500

Capture the flag



Capture the flag

- CTFs are team-based cybersecurity competitions, often involving real-world attacks
- You score points by *capturing the flag* of a given challenge, where the flag is a secret string hidden from you

Capture the flag

Challenges are divided into many categories:

Capture the flag

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- **Pwn**: Exploit binaries or services, bypass security measures, pop shells in random people's servers.

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- **Reverse**: Break down and understand the logic of a binary. Think about old-school keygens or anything related to malware analysis.

Capture the flag

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- **Reverse**: Break down and understand the logic of a binary. Think about old-school keygens or anything related to malware analysis.
- **Stegano, OSINT, Misc**: Join the team for more information :)

CTF is like football



CTF is like football

- Effective age: 14-35 yrs old



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- Play on weekends



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- International mega tournaments



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- Physically exhausting



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- Mega teams like real madrid



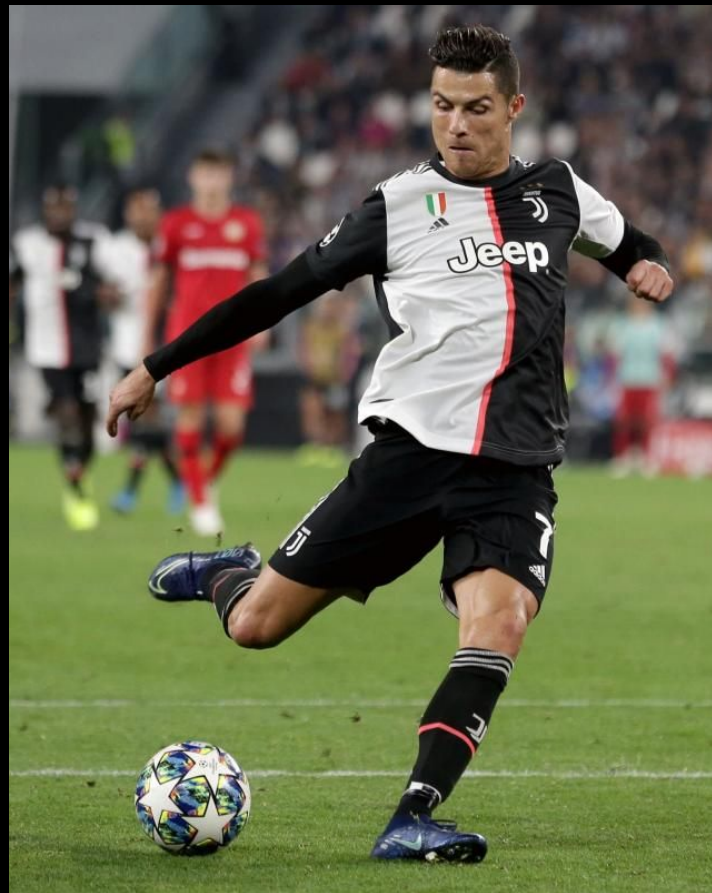
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- Buying/selling of players



CTF is like football

- Effective age: 14-35 yrs old
- Play on weekends
- International mega tournaments
- Physically exhausting
- Mega teams like real madrid
- Buying/selling of players
- Doping



Doping



Doping

- Somehow still legal
- Good taste
- Sparkling
- Way too much sugar
- Goes directly to your brain
- No more sleep
- No more hunger
- Just focus on flagging



Skills tested

Skills tested

- Competitive programming

Gennady Korotkevich



Korotkevich in 2014

Born	September 23, 1994 (age 29) Gomel
Nationality	 Belarus
Other names	Codeforces handle: tourist
Education	ITMO University
Known for	Programming prodigy; highly ranked sport programmer from an early age
Awards	Codeforces peak rating 3979 (22 January 2022)

Skills tested

- Competitive programming
- Debugging



PS PlumbingSupply.com
How to Unclog a Toilet

Skills tested

- Competitive programming
- Debugging
- Assembly

```
section      .text
global      _start                      ;must be declared for linker (ld)

_start:                                          ;tell linker entry point

    mov     edx,len                      ;message length
    mov     ecx,msg                      ;message to write
    mov     ebx,1                        ;file descriptor (stdout)
    mov     eax,4                        ;system call number (sys_write)
    int     0x80                         ;call kernel

    mov     eax,1                        ;system call number (sys_exit)
    int     0x80                         ;call kernel

section      .data

msg         db  'Hello, world!',0xa        ;our dear string
len         equ $ - msg                   ;length of our dear string
```

Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins



Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- ripgrep
- fd
- tmux
- bash for loops
- jq
- ssh
- man

Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- Reading cursed crypto papers



Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- Reading cursed crypto papers
- Guessing

It's 3am on the last night, the pwn players have all gathered around one screen, there is no leak, they're just trying to guess ASLR. 2006 [colorized]



Team rating

2022 2021 2020 2019 2018 2017 2016 2015 2014 2013
2012 2011

Place	Team	Country	Rating
👑 1	Project Sekai		228.454
2	r3kapig		169.606
3	u0K++		157.573
4	idek		140.009
5	WreckTheLine		126.752
6	C4T BuT S4D		118.427
7	Never Stop Exploiting		117.770
8	DiceGang		112.467
9	Chaitin Tech		109.760
10	perfect r00t		109.760

[Full rating](#) | [Rating formula](#)

Upcoming events 📅 🔊

Open [Finals](#)

Past events 🔊

With scoreboard [All](#)

🏠 Ugra CTF Quals 2022

March 01, 2022 07:00 UTC | On-line | [Weight voting in progress](#)

Place	Team	Country	Points
👑 1	s3qu3nc3		62.000
2	Project Sekai		45.988
3	FaKappa		37.234

[103 teams total](#) | [Tasks and writeups](#)

🏠 wtfctf 2022

Feb. 28, 2022 18:29 UTC | On-line | [Weight voting in progress](#)

Place	Team	Country	Points
👑 1	Project Sekai		38.420
2	1nf1n1ty		27.007
3	Flag Poisoning		22.772

[246 teams total](#) | [Tasks and writeups](#)

🏠 TSJ CTF 2022

Feb. 28, 2022 05:00 UTC | On-line | [Weight voting in progress](#)

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

flagbot



<https://flagbot.ch>

- “The grass is always greener on the other side”
- Academic ETH team

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

flagbot



<https://flagbot.ch>

- “The grass is always greener on the other side”
- Academic ETH team

organizers



<https://org.anize.rs/>

- Born from a meme
- very strong
- polygl0ts + flagbot + cr0wn + some small other teams

The teams

polyglots



Lausanne Sport

- Play for fun
- Fair play
- Learn together
- Cute small local team

The teams

polygl0ts



Lausanne Sport

- Play for fun
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flagbot



F.C. Zurich

- Sister team
- Currently in a lesbian relationship with us
- Pretend to compete but it's just foreplay

The teams

polygl0ts



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- Play for fun
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flagbot



F.C. Zurich

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organizers



Real Madrid

- Play to win
- Top tier international finals
- Tryhard behaviour (doping, swearing, becoming furry, etc.)
- Cursed team, constant drama

Organizers very strong

#1 in the world two years ago

Always participating in DEFCON FINALS

		2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
		2012	2011								
Place	Team	Country		Rating							
👑 1	perfect blue			1409.969							
2	organizers			1173.314							
3	Super Guesser			1036.950							
4	Balsn			1010.773							
5	DiceGang			951.334							
6	More Smoked Leet Chicken			901.412							
7	Bushwhackers			819.242							
8	Plaid Parliament of Pwning			808.852							
9	Never Stop Exploiting			794.366							
10	Nu1L			776.762							

[Full rating](#) | [Rating formula](#)

Upcoming events >

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2023

2022

2021

2020

2019

2018

2017

2016

2015

2014

2013

2012

2011

Place	Team	Country	Rating
👑 1	organizers		1511.323
2	perfect r00t		1283.824
3	Never Stop Exploiting		1025.931
4	C4T BuT S4D		1013.462
5	Water Paddler		942.923
6	Nu1L		903.930
7	justCatTheFish		887.498
8	Bushwhackers		817.228
9	DiceGang		756.228
10	kalmarunionen		737.250

[Full rating](#) | [Rating formula](#)

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Took part in many top-tier CTFs:

- ProCTF (Abu Dhabi)

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
2013	2012	2011							
Place	Team	Country	Rating						
👑 1	Blue Water		1272.461						
2	kalmarunionen		1030.562						
3	C4T BuT S4D		1010.123						
4	justCatTheFish		887.259						
5	r3kapig		822.413						
6	SKSD		761.927						
7	organizers		708.467						
8	thehackerscrew		698.748						
9	Project Sekai		684.336						
10	WreckTheLine		677.514						

[Full rating](#) | [Rating formula](#)

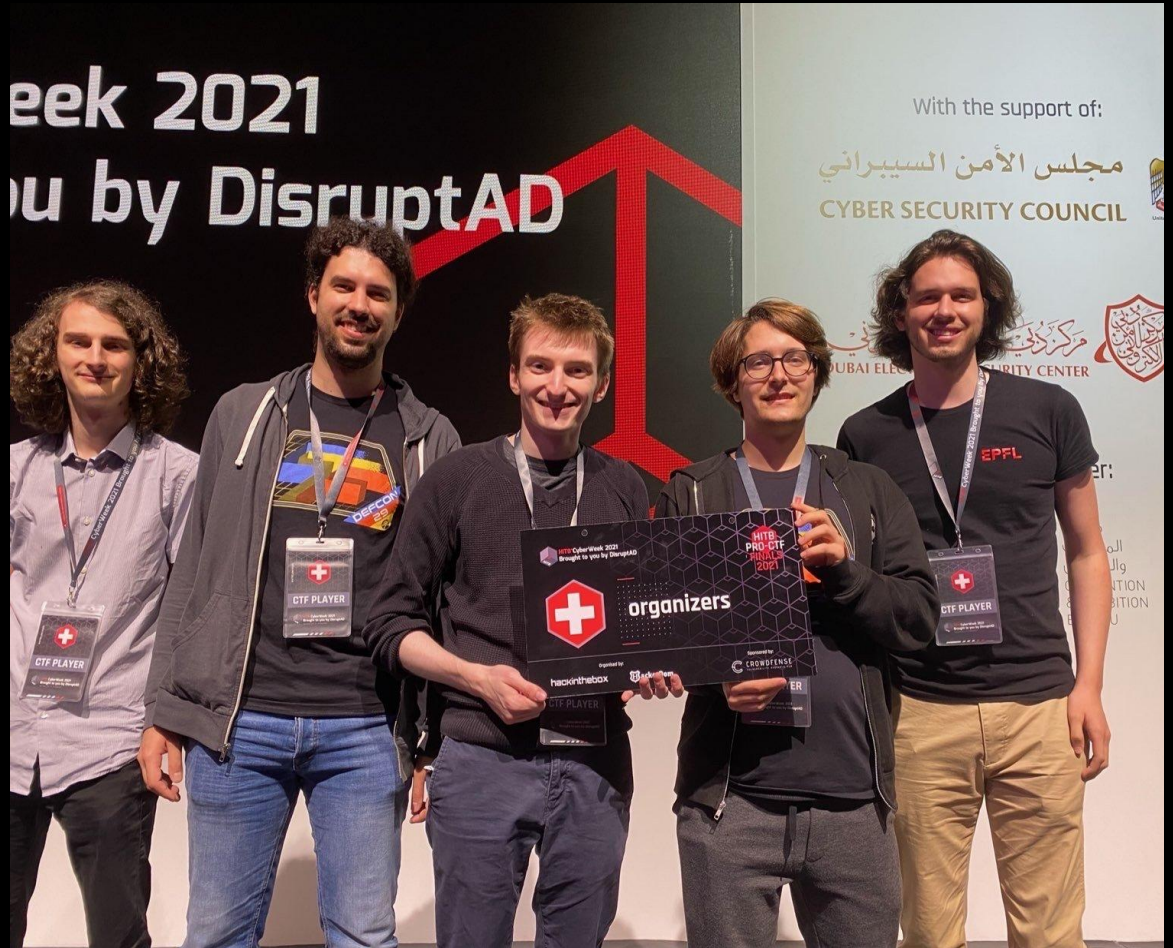
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Always participating in DEFCON FINALS

Took part in many top-tier CTFs:

- ProCTF (Abu Dhabi)
- DEFCON FINALS (Las Vegas)



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- Google CTF (London, Tokyo)



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- Google CTF (London, Tokyo)
- Midnight Sun (Stockholm)



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- Midnight Sun (Stockholm)
- XCTF (Chengdu, China)



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- XCTF (Chengdu, China)
- MOLECON (Torino)
- SECCON (Seoul)



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- XCTF (Chengdu, China)
- MOLECON (Torino)
- SECCON (Seoul)
- HXP CTF (Hamburg)
- HITCON
- Midnight Sun CTF (Stockholm)
- HackASat (Las Vegas)
- Insomnihack (Geneva)



How to git gud

- CTFs are hard, very involved
- Easy to get discouraged

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How to get better:

- Play with people of your level

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- Play with people above your level

How to git gud

- CTFs are hard, very involved
- Easy to get discouraged

How to get better:

- Play with people of your level
- Play with people above your level
- Read writeups

Writeups

> Very often it will happen that a chall is too hard for you.

Take time after the CTF and read a writeup!

Writeups

> Very often it will happen that a chall is too hard for you.

Take time after the CTF and read a writeup!

> Sometimes you'll be able to solve a challenge.

Write a writeup!!!

Tools

Tools

- ghidra



Tools

- ghidra/IDA



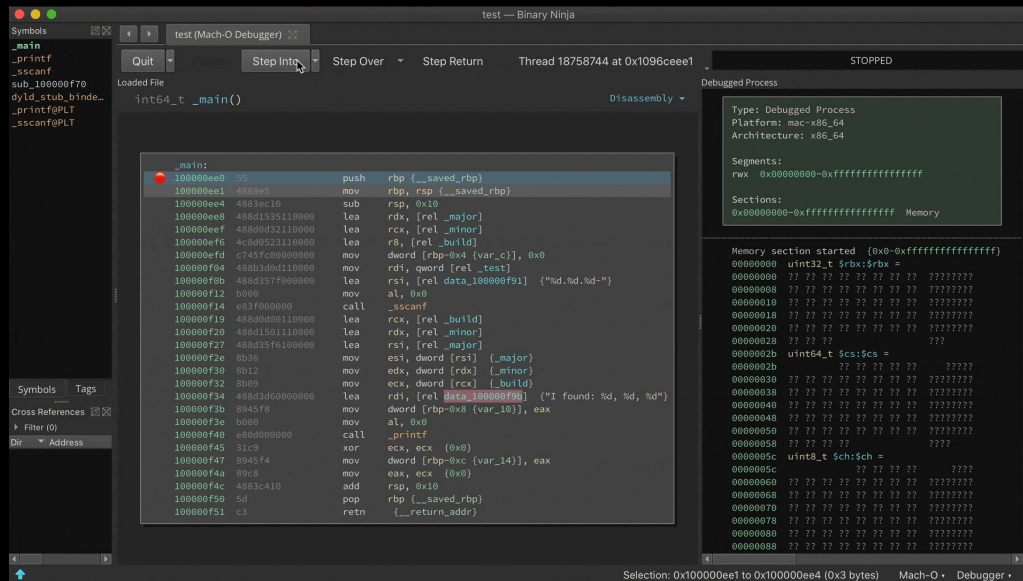
Tools

- ghidra/IDA/binja



Tools

- ghidra/IDA/binja



Tools

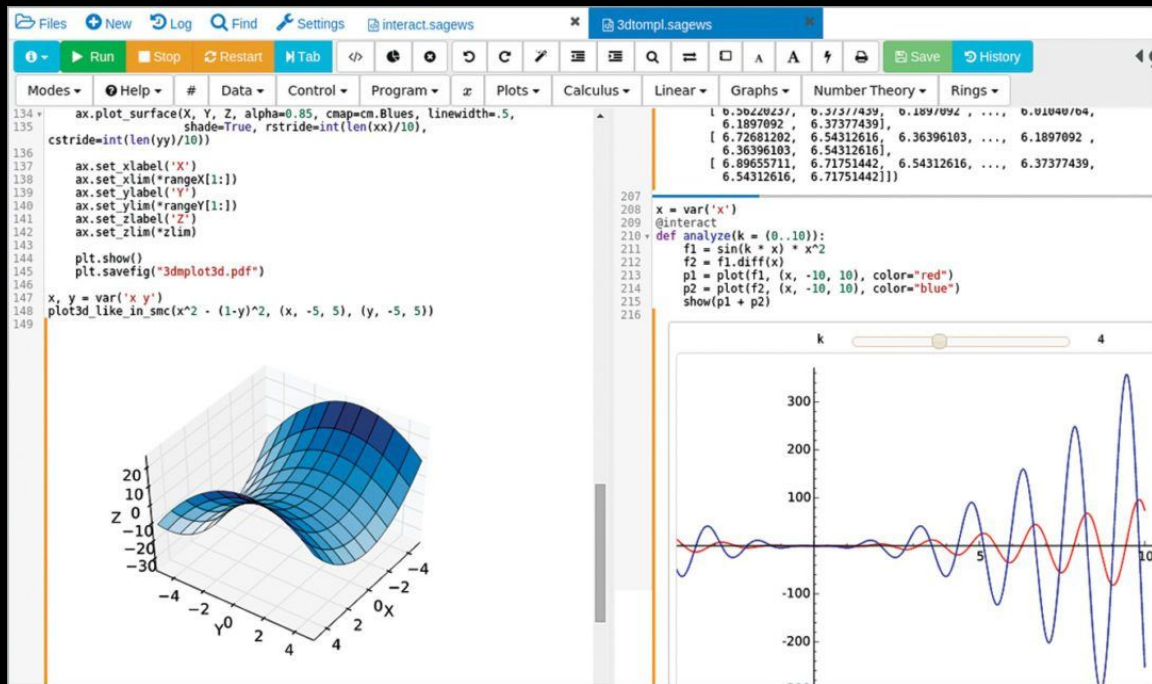
- ghidra/IDA/binja
- pwntools



```
from pwn import *  
r = remote("127.22.12.3", 8000)  
r.sendline(asm(shellcraft.sh()))  
r.interactive()
```

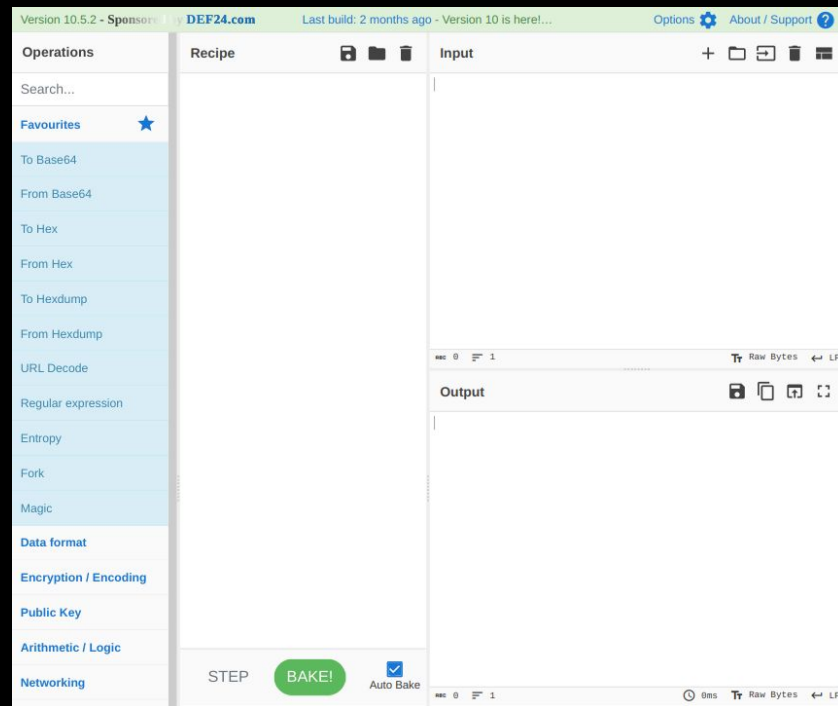
Tools

- ghidra/IDA/binja
- pwntools
- sage



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3
- gdb

```

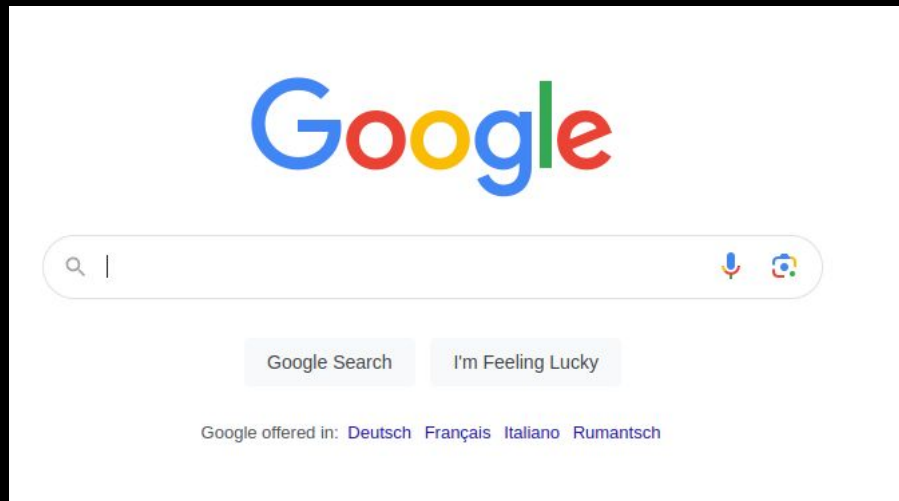
-----
pwndbg> b *main
Breakpoint 1 at 0x8048deb
pwndbg> r
Starting program: /data/crackme

Breakpoint 1, 0x8048deb in main ()
LEGEND: STACK | HEAP | CODE | DATA | RWX | RODATA
-----
REGISTERS
EAX 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
EBX 0x80481d0 (_init) <-- push ebx
ECX 0x3ea8d01
EDX 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
EDI 0x0
ESI 0xb21a00c (_GLOBAL_OFFSET_TABLE_+12) --> 0xb076c10 (_rawmemchr_uses2_bef) <-- mov ecx, dword ptr [esp + 4]
EBP 0x0
ESP 0xffffd48c --> 0x80492e7 (_libc_start_main+487) <-- add esp, 0x10
EIP 0x8048deb (main) <-- lea ecx, [esp + 4]
-----
CODE
=> 0x8048deb <main>      lea    ecx, [esp + 4]
0x8048def <main+4>      and    esp, 0xffffffff
0x8048df2 <main+7>      push   dword ptr [ecx - 4]
0x8048df5 <main+10>     push   ebp
0x8048df6 <main+11>     mov    ebp, esp
0x8048df8 <main+13>     push   esi
0x8048df9 <main+14>     push   ebx
0x8048dfa <main+15>     push   ecx
0x8048dfb <main+16>     sub    esp, 0x4c
0x8048dfe <main+19>     mov    ebx, ecx
0x8048e00 <main+21>     cmp    dword ptr [ebx], 1
-----
STACK
00:0000| esp      0xffffd48c --> 0x80492e7 (_libc_start_main+487) <-- add esp, 0x10
01:0004|          0xffffd490 <-- 0x1
02:0008|          0xffffd494 --> 0xffffd504 --> 0xffffd67f <-- '/data/crackme'
03:000c|          0xffffd498 --> 0xffffd50c --> 0xffffd68d <-- 'XDG_VTNR=2'
04:0010|          0xffffd49c --> 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
05:0014|          0xffffd4a0 <-- 0x0
06:0018| eax edx  0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
07:001c|          0xffffd4a8 --> 0xb21a00c (_GLOBAL_OFFSET_TABLE_+12) --> 0xb076c10 (_rawmemchr_uses2_bef) <-- mov ecx, dword ptr [esp + 4]
-----
BACKTRACE
> f 0 8048deb main
f 1 80492e7 _libc_start_main+487
Breakpoint *main

```

Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3
- gdb
- google



LakeCTF

CTF we are hosting

December 14-15 quals

Finals in May (Balelec
afterparty!)

[call for challenges!](#)



Starts in: 37d 19h 39m 35s

LakeCTF is a jeopardy-style security hacking contest organized by the [polygl0ts](#) CTF team (plus some friends!) and the [EPFL School of Computer and Communication Sciences \(IC\)](#).

You must first compete in an online qualification round, and then physically compete in the finals in Lausanne. The top teams will have the unique chance to meet the top members of the EPFL research community.

We wish you the best of luck !

> Qualls schedule

Qualification start	Sat 4th of November, 18:00 UTC
Qualification end	Sun 5th of November, 18:00 UTC

> Finals schedule

Fri 27th of April 2024	Arrival Tour of Lausanne
Sat 28th of April 2024	9:00 CTF Finals Start

Recap

- Website: `polygl0ts.ch`
- Discord! (link in website)
- Join mailing list (send empty mail to ctf-subscribe@listes.epfl.ch)
- CTFs usually on weekends
- Weekly tutorials on Fri 17:00 INF1, for newcomers



YOUR FIRST CTF

CTF organized by our friends
at ETH.

For beginners

We go on Saturday morning
until Saturday evening

Travel sponsored by polygl0ts!



björnCTF



> A ~~Cybercrime~~ Hacking Competition

Friday 04. October
13:00 - Sunday 13:00
STUZ CAB F21

team_size ≤ 4

Beginners welcome.
<https://björnc.tf>

> _